Dimitrios Fotiou

Computer Vision & Perception Engineer

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github.com/dimfot3 — linkedin.com/in/dimitris-fotiou — Website

Professional Summary

Computer Vision and Perception Engineer specializing in real-time, multi-sensor perception 2D or 3D sensors and deep learning deployment optimized for production. Proven across autonomous vehicles, robotics, industrial inspection, and disaster response, with research and industry experience.

Technical Skills

Programming: Python, C/C++, SQL, MATLAB

Machine Learning / Computer Vision: PyTorch, TensorFlow, ONNX Runtime, TensorRT, CUDA, Nvidia Triton, image segmentation, object detection, test-time adaptation, vision foundation models

Robotics / Sensors: ROS2, Gazebo, CARLA, LiDAR, RGB cameras, IR cameras, IMU, GPS

Software / Tools: Linux, Git, Docker, Flask API, MinIO, Context Broker, CVAT, FiftyOne, Weight&Biases

Parallel Computing: CUDA, MPI, OpenMP

Work Experience

Software Engineer – Machine Learning Sep 2025 – Present
D-Cube *Thessaloniki, Greece*

- Proposed and developed a centralized foundation model for aluminium defect classification, replacing per-client models and legacy classifiers with a Retrieval-Augmented classification, enabling zero-shot classification for new clients with minimal deployment effort.
- Reduced latency and memory consumption using TensorRT and Nvidia Triton.
- Built 3D computer vision solutions for defect detection with Intel Real Sense L515, Sick PicoScan 150
- 6D zero-shot profile detection using FoundationPose optimized with TRT for VR application (POC)

Software Engineer – Machine Learning Aug 2024 – Sep 2025
AIIA.CVML *Thessaloniki, Greece*

- Developed dockerized real-time natural disaster detection systems using RGB and IR camera data for TEMA project.
- Integrated MinIO, Context Broker, and ROS2 pipelines for real-time data processing.
- Improved wildfire detection accuracy by over 5% on challenging state-of-the-art dataset FLAME2 and tested successfully on actual fire in Sardinia without explicit fine tune on that area.
- Teaching assistant for the MSc course *Autonomous System Perception* and Lecturer at AIDA Summer School 2025 on *Wildfire Image Analysis*.

Programmer and Network Security Analyst Oct 2023 – Jul 2024
Hellenic Army *Greece*

- Maintained and secured a 150+ node hierarchical network using Active Directory and Windows Server.
- Performed software upgrades, malware analysis, and system maintenance.
- Developed and maintained internal websites using HTML, CSS, and JavaScript and scrapping apps.

Software Engineer – Machine Learning Intern Mar 2022 – Sep 2022
Information Technologies Institute, CERTH *Thessaloniki, Greece*

- Developed LiDAR-based loop closure detection methods (Scan Context based) for automotive SLAM applications.
- Evaluated algorithms on both real-world and simulated CARLA datasets.

Projects

Autonomous Formula Student Vehicle – ARISTURTLE Aug 2020 – Oct 2021
Software Engineer

- Developed perception modules for Greece's first Formula Student autonomous vehicle, Thetis.
- Implemented cone detection and 6D pose estimation using YOLO and Perspective-n-Points.
- Optimized inference using CUDA and TensorRT and quantization.
- Applied Kalman-based sensor fusion using GPS, IMU, hall, and resolver sensors.
- Built a realistic 3D vehicle dynamics simulation in Gazebo with integrated perception and localization sensors.
- Achieved 3rd place in Skidpad and 4th place in Dynamic Events at FS East, Hungaroring.

Additional repositories (HPC/parallel computing, robotics simulation, computer vision) at github.com/dimfot3.

Education

MSc Electrical and Computer Engineering (Integrated Master, 5-year) Oct 2018 – Jun 2023
Aristotle University of Thessaloniki, Faculty of Engineering *Greece*

- Graduated with distinction, GPA: 8.99/10.
- Received scholarships for excellence in mathematics, programming, and engineering.
- Diploma thesis: "Indoor Environment Replication and Human Detection using LiDAR Technology", developed a robust plane-detection algorithm and trained PointNet/PointNet++ models for real-time human detection and pose estimation on 3D LiDAR point clouds, validated in ROS2/Gazebo/Blender simulations.

Publications

- D. Fotiou, V. Mygdalis, and I. Pitas, "RoboFireFuseNet: Robust Fusion of Visible and Infrared Wildfire Imaging for Real-Time Flame and Smoke Segmentation," *Pattern Recognition Letters*, 2026.
- Dimitrios Fotiou, Vasileios Mygdalis, and Ioannis Pitas, "TestMate: Test-Time Domain Adaptation Aided by Lightweight Vision Foundation Model," *arXiv preprint arXiv:2607.03810, 2026*

Languages

Greek: Native — **English:** C2